MILITARY SPECIFICATION

STARTER-GENERATOR ENGINE STU-6A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aircraft gas turbine engine driven starter generators for use in 28-volts (V) direct current (dc) systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-E-489	Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces)
TT-S-1732	Sealing, Pipe Joint and Thread, Lead Free, General Purpose
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-704	Treatment and Painting of Material
DOD-D-1000	Drawing, Engineering And Associated List
MIL-M-3171	Magnesium Alloy, Processes For Pretreatment And Prevention of Corrosion on
MIL-S-5002	Surface Treatments And Inorganic Coatings For Metal Surfaces Of Weapon Systems
MIL-C-5541	Chemical Conversion Coatings On Aluminum Alloys
MIL-C-6021	Castings, Classification And Inspection of

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MIL-R-6809	Regulator, Voltage, 30-Volt, Direct Current Generator, General Specification For
MIL-S-7742	Screw Threads, Standards, Optimum Selected Series, General Specification For
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-A-8625	Anodic Loadings, for Aluminum And Aluminum Alloys
MIL-R-23761	Regulator, Voltage, And Control Panels Aircraft, Direct Current Generator, General Specification For

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specification And Standards Order Of Precedence For The Selection Of
MIL-STD-147	Palletized Unit Loads
MIL-STD-794	Parts And Equipment Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-826	Electromagnetic Interference Test Requirements And Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STI)-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Water-proofing; With Appropriate Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

AIR FORCE - NAVY AERONAUTICAL

AND10262	Flange, Type XII Accessory Mounting
AND20002	Drive-Type XII Engine Accessory

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3915-82	Packaging, Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The starter-generator furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no standard parts. In any case. commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor’s drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 MATERIALS

3.3.1 Fungus proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

* 3.3.2.1 Dissimilar metals. Unless protected against electrolytic corrosion. dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STP-889.

3.3.3 Castings. Castings shall be clean, sound, and free from cracks, excessive porosity and other defects. Castings shall be classified in accordance with MIL-C-6021, and shall be so indicated on the applicable detail drawings.

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3.4 Design and construction. The starter-generator shall be so designed and constructed that no parts will work loose in service and shall be built to Withstand the strains. jars, vibrations, and other conditions incident to shipping, storage, Installation, and service. The starter-generator shall be so designed that no part can be assembled in reverse position and cause faulty operation or lead to eventual failure.

3.4.1 Simplication. Simplicity of design resulting from the use of the same part for as many applications as possible in a given model is highly desirable. (For example: Use of identical ball bearing assemblies on both ends of the armature.)

3.4.2 Starter. The starter shall be designed to operate with series and shunt fields connected. The minimum torque delivery of the starter shall be as specified on Figures 1 and 2 under the conditions specified thereon.

3.4.3 Cenerator. The generator shall be designed to operate with a voltage regulator in accordance with MIL-R-6809 or MIL-R-23761. The generator shall be connected short shunt.

3.4.3.1 Generator field. The generator shunt field shall be so designed that the maximum regulated field current shall be not more than 8 amperes (amp) and the power to be dissipated by the regulator shall be not more than 90 watts (w). Accidental closure of the reverse-current relay shall not reverse the field.

3.4.4 Mounting flange. The starter-generator mounting flange shall conform to AND10262 with slotted mounting holes as specified thereon.

3.4.5 Torsional vibration damper. Provisions shall be incorporated in the generator for installation of a torsional vibration damper between the drive spline and armature. The vibration damper shall not be furnished unless specified by the qualifying activity (see 6.2).

3.4.6 Bearings. The armature in the starter-generator shall be supported on both ends by sealed ball bearings. The bearing support on the drive-shaft end shall be vented to equalize pressure or suction on both sides of the bearing. The method of accomplishing this requirement shall meet the approval of the qualifying activity. Ball bearings used shall incorporate contact seals.

3.4.7 Terminal parts. The terminal block shall be provided with three 3/8-24 studs with UNF-2A threads and two No. 10-32 studs with UNF-3A threads. Each stud shall be provided with a plain washer and self-locking nut. Terminals shall be as shown in Table I. The spacing of the insulating barriers and 3/8-24 terminal studs shall be arranged to provide for the installation of terminal lugs having a maximum width of 0.800 inch. The design of the terminal block shall not depend on current to be transmitted by the studs.

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3.4.7.1 Terminal marking. The terminals shall be located and marked as shown on Figure 3.

TABLE I. Terminal Identification

	Terminal Size	Marking
Shunt field positive	(No. 10)	A+
Generator, starter, and shunt field negative	(3/8)	E-
Generator positive	(3/8)	B+
Starter winding positive	(3/8)	C+
Paralleling	(No. 10)	D

3.4.8 Brushes.

3.4.8.1 Brush accessibility. Brushes shall be accessible for inspection and replacement when brush cover bands are removed (see Figure 3).

3.4.8.2 Brushfit. The face of each brush shall contact its commutator 100 per cent in the direction of rotation and for at least 75 per cent of brush dimension parallel to the shaft. There shall be no evidence of grooving or other surface damage to the face of the brush.

3.4.8.3 Cartridge type. The brushes in cartridge-type brush assemblies shall not rotate when mounted.

3.4.8.4 Spare brushes. All spare brushes shall be formed to a radius equal to the radius of the new commutator and shall be identical to the corresponding brushes approved by the qualification tests.

3.4.8.5 Brush position. The brush rigging shall be fixed position (non-shifting) type. The brush rigging position shall be identical for all generators of the same model number made by the same manufacturer.

3.4.8.6 Brush marking. Each brush shall have a diagonal groove in the anti-drive side, from the contact surface to a point on the front or back of the brush, which indicates 100 per cent of allowable wear.

3.4.8.7 Brush type. Brushes shall be of the immediate filming type which do not require prolonged operation of the starter-generator to establish an altitude protective film on the commutator. Rivets used to secure the leads to the brushes shall be silver soldered to the pigtail at point of riveted contact.

* 3.4.9 Cooling. For generator operation, cooling air shall be provided through a 2 inch diameter duct which shall not be a part of the starter-generator. The starter-generator shall include provisions for entrance of external air for cooling purposes through the anti-drive end, or brush access windows. When cooling air is required to enter through the anti-drive end,

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the brush cover will be installed; when cooling air is required to enter through the brush access windows, a cover will be installed on the anti-drive end to prevent the loss of cooling air. The starter-generator will be furnished with the Air Inlet Cover on the anti-drive end and the Brush Cover installed. See Figure 3.

3.4.10 Shaft deflection. The starter-generator shall be designed to operate with the driveshaft misaligned to the maximum allowable limits defined by the tolerances of the mating engine accessory drive, AND20002, type XII-D.

3.4.11 Operating attitude. The starter-generator shall be designed to operate in any installed position.

3.4.12 Rotation. The direction of rotation of the splined drive shall be as specified on Figure 3.

3.4.13 Overhung moment. The overhung moment shall not exceed 200 inch-pounds.

3.5 Reliability. The starter-generator shall be designed and constructed to provide an inherent reliability of 1,000 hours maintenance-free operation under the conditions required herein except for one change of brushes. To meet this requirement, careful attention shall be given to selection of materials, treatment of materials, finishes, securing of all electrical conductors, and the assembly of each and every part.

3.6 Maintainability. The starter-generator shall have a minimum number of parts consistent with reliability. Its designs shall permit easy assembly, disassembly, location of trouble sources, and maintenance with tools and equipment normally available commercially.

3.6.1 Armature. The starter-generator shall be so constructed that the armature is readily removable for maintenance with minimum disassembly. In no case shall it be necessary to disconnect soldered or brazed connections.

3.6.2 Commutator. A turned down section (step) shall be provided at the outboard end of the commutator, the diameter of which shall be the minimum diameter to which the commutator can be safely reduced during its maintenance life. The minimum diameter shall be consistent with brush stability.

3.7 Performance.

3.7.1 Environmental conditions. The starter-generator shall be capable of satisfactory operation under the following environmental conditions:

a. Ambient temperatures ranging from -54° Centigrade (C) to **+71°C.**

b. Relative humidity up to 100 per cent, including conditions wherein condensation takes place in the form of both water and frost.

c. Vibration incident to service use.

d. Fungus growth as encountered in tropical climates.

e. Operation in an explosive vapor within or surrounding the equipment.

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3.7.1.1 Low and high temperature storage. The starter-generator shall operate satisfactorily after having been stored at temperatures ranging from -62°C to +93°C.

3.7.2 Maximum speed for regulation. The starter-generator, when operated as a generator, shall be capable of delivering rated voltage at no-load, at not less than 9,000 revolutions per minute (rpm) with no more than 35 ohms in the external field circuit. (See 4.6.2)

3.7.3 Minimum speed for regulation. The generator shall be capable of delivering full-rated load at rated voltage, continuously, at the minimum speed for regulation (4,400 rpm) with an external resistance in the shunt-field circuit of 1.25 ohms minimum and not more than 8 amp field current. At 3,800 rpm the generator shall be capable of delivering 75 per cent rated load at 26V, using a voltage regulator conforming to MIL-R-6809 or MIL-R-23761.

3.7.4 Heating commutation and minimum speed. The generator shall be capable of meeting the operating conditions specified in 4.6.4. The required resistance external to the shunt field when operating at the minimum speed, while the generator is hot, shall be not less than 1.25 ohms. (See 4.6.4)

3.7.5 Overspeed. The starter-generator shall be capable of being driven at a driveshaft speed of 10,000 rpm with no-load for 5 minutes without mechanical failure or impairing of electrical performance. (See 4.6.5)

3.7.6 No-load speed. When operating as a starter at no-load and with 23 volts applied to the appropriate terminals, the starter-generator shall develop a minimum driveshaft speed of 2,500 rpm.

3.7.7 Radial vibration. The radial vibrations shall not exceed 0.001 inch total deflection at maximum rated speed when operated as a starter. (See 4.6.7)

3.7.8 Paralleling voltage drop. Under conditions of 200 amp, 30V dc, 8,000 rpm with starter-generator stabilized in temperature and an air inlet temperature of 25°C, the voltage drop between terminals D and E shall be set at 2V - 5 per cent. The generator equalizing voltage shall be obtained between terminal E and the negative brushes or a tap in the winding between negative brushes and terminal E. (See 4.6.8)

3.7.9 Dielectric strength. The starter-generator shall withstand a test voltage of 500V for 1 minute, or 600v for 1 second, at commercial frequency applied between windings and between each winding and frame. (See 4.6.9)

3.7.10 Ripple voltage-generator. The starter generator ripple voltage, when instrumented as shown on Figure 4 and operated at speeds of 3,800, 6,000 and 8,100 rpm under no-load, half-load, and rated-load, shall not exceed 1.5V.

3.7.11 Cranking torque and speed. The starter-generator, when connected and operated as a starter, shall be capable of delivering the minimum cranking torque and speed requirements specified on Figures 1 and 2, with an external resistance of 1.25 ohms corrected in the shunt field circuit. (See 4.6.11)

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3.7.12 Endurance-starter. The starter-generator, when operated as a starter, shall be capable of completing a 600-cycle endurance test as specified in 4.6.13 without resultant damage to or replacement of component parts.

3.7.13 Efficiency-generator. The generator shall be capable of attaining an efficiency of not less than 60 per cent at rated terminal voltage and at the speed and load conditions specified in 4.6.14.

3.7.14 Endurance-generator. The generator shall be capable of performing eight cycles of runs 1, 2 and 3, as shown in Table II, followed by a single run, No. 4.

TABLE II. Endurance - Generator

Run No.	Time (hrs)	Speed (rpm)	Current (amp)	Voltage (volts)	Altitude (feet)	Cooling Air	
						Inlet Temp °C	Press drop across unit inches H_2O
1	¼	3,800	150	26	0	24 to 40	6.0
2	¼	8,100	200	30	0	24 to 40	12.0
3	6	8,000	165	30	60,000	*-50 to -60	6.0
4	100	8,000	200	30	0	24 to 40	12.0

*Dew point - 75°C ± 5°C

3.7.15 Consecutive cycle-starter. The starter-generator shall be capable of performing three cycles of operation at 30-second intervals. Each cycle shall consist of three consecutive phases as shown in Table III. No external cooling shall be permitted for this test.

TABLE III. Endurance - Starter

MAX TERMINAL POTENTIAL - 28 VOLTS

Phase A - This phase will consist of accelerating a flywheel having a 50 lb-ft^2 polar moment of inertia from rest to 500 rpm within 3 seconds.
Phase B - As soon as the speed designated in phase A is attained, a torque load of 45 lb-ft, ±2, shall be applied for a 3-second period.
Phase C - At the end of phase B the torque shall be removed and the starter allowed to accelerate the flywheel to 2500 rpm within 21 seconds. External cooling may be used to expedite testing and to prevent excessive temperature. A speed less than 2500 rpm cut-out may be utilized provided the total horse-power-second output of the starter is equivalent to that required to accelerate a frictionless flywheel to the specified cut-out speed of 2500 rpm including phase B loading.

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3.7.16 Sea water immersion. The starter-generator shall withstand five sea water immersion cycles without replacement of parts except for brushes and bearings. The sea water shall conform to the salt solution specified in MIL-STD-810 except that 20 ± 2 parts by weight of salt shall read 4 ± .1 and the specific gravity shall be 1.026 to 1.034. The cycles shall conform to the following:

a. Complete immersion in sea water for 2 hours.

b. Removal from sea water and allowed to stand for 2 hours at normal room ambient conditions.

c. Complete submersion in fresh water for 5 minutes.

d. Removal from fresh water and allowed to stand for 3-1/2 hours at normal room ambient conditions.

Disassembly, decontamination of components, followed by not more than 16 hours of drying at a temperature of 121 ± 10°C.

f. Reassemble with new brushes followed by operation for a period of 1 hour at 8,000 rpm, 200 amp, 30V dc at normal room ambient with cooling air pressure drop across the generator equivalent to 12 inches of water.

3.7.17 Drive shear section. A shear section shall be incorporated immediately after the drive spline. Shear shall occur between 880 to 1,000 inch (in) pound (lb). The shear section shall be designed so that its failure will not result in loose pieces which could damage adjacent components. In addition, replacement of a failed shear section shall be possible without requiring a major disassembly of the starter-generator.

3.7.18 Bearing failure. The starter-generator, minus the seals, cage, and lubricant in either bearing shall not become a potential fire hazard when operated under conditions of 200 amp, 30V dc, 8,000 rpm.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.9 Radio noise interference. When operating as a generator, the starter-generator shall be provided with the necessary radio noise suppression features to suppress its radio noise to the limits specified in 4.6.12.1. These features shall include adequate filters and enclosing case construction which will prevent noise leakage through joints, seams, and mating surfaces. The volume and weight of filtering equipment required shall be minimized by the application of proper electrical and mechanical design and construction. The filtering equipment shall be located inside the starter-generator.

3.9.1 Bonding. The frame structure of the starter-generator shall be so assembled that there will be a continuous, low impedance path from the commutator end bell to face of mounting flange. The direct-current resistance between commutator end bell and mounting flange shall be not greater than 0.001 ohm. The area normally covered by starter-generator mounting nuts shall be free from insulating type of surface treatment provided the 0.001 ohm frame structure resistance is met. (See 4.6.12.1)

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3.10 Dimensions. The starter-generator shall conform to the dimensions shown on Figure 3.

3.11 Weight. The starter-generator, shall be as light as practicable, consistent with the requirements for performance, safety, and reliability, and shall weigh not more than 41 pounds.

3.12 Finishes and protective coatings.

3.12.1 Cleaning, treating, and painting. Unless otherwise specified herein, metal parts shall be cleaned, treated, and painted in accordance with MIL-T-704. However, when plating is specified, the-cleaning of the metal shall conform to MIL-S-5002.

3.12.2 Cadmium plating (steel parts). With the exception of the parts listed below, all exterior steel parts, and other steel parts subject to corrosion and not in constant contact with oil, shall be cadmium plated in accordance with QQ-P-416 type II or III and Class 2 as applicable. In cases where cadmium plating is considered impractical or undesirable an equivalent treatment which is acceptable to the procuring activity shall be provided.

a. Corrosion-resistant steel parts.

b. Members or portions of members which act as bearing of journals.

c. Magnet frame.

d. Steel end-bells, brush cover band.

3.12.3 Anodic film for aluminum and aluminum parts. Aluminum and aluminum alloy parts shall be covered by an anodic film conforming to MIL-A-8625.

3.12.3.1 The aluminum oxide film deposited by the treatment shall be removed from actual contact area of all surfaces required to act as a path for electric current and from local areas under screws, nuts, et cetera, used for assembly or mounting purposes, to provide an adequate bonding connection.

3.12.3.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical coat conforming to MIL-C-5541.

3.12.3.3 Where primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.12.3.4 Castings containing nonaluminum alloys integral inserts may be treated with chemical films in accordance with MIL-C-5541 in lieu of anodizing.

3.12.3.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.12.4 Magnesium-alloy parts. All magnesium-alloy parts shall be surface treated for protection against corrosion in accordance with MIL-M-3171.

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* 3.12.5 Paint finishes. All exposed metal surfaces shall be painted with one coat of primer conforming to MIL-P-8585 and two finish coats of enamel conforming to TT-E-489, type I, or one coat of type II enamel, except as follows :

a. Corrosion-resistant steel, brass, copper or bronze parts.

b. Cable.

c. Working surface.

d. Threads.

e. Oil holes.

f. Cadmium - or zinc-plated parts.

g. Those parts on which the application of paint has been demonstrated to the satisfaction of the procuring activity to be impractical or unnecessary.

3.13 Screw threads. Unless otherwise specified, the threads of all machine screws 0.06 inch in diameter, or larger, shall conform to MIL-S-7742.

* 3.13.1 Coating threaded parts. All threaded parts, except electrical terminals and safety stop nuts, shall have a thin coating of compound conforming to TT-S-1732 applied to the threads. No more of this compound shall be applied than is necessary to cover the surface of the threads.

3.14 Lubrication. Where lubrication is required, it shall be in accordance with MIL-STD-838. The lubrication shall be such that it will insure satisfactory operation when the starter-generator is permanently mounted in any position. The starter-generator shall not require added lubrication during 1,000 hours of operation. The lubrication of the starter-generator shall be independent of the engine lubrication system.

3.15 Altitude rating chart. The starter-generator manufacturer shall supply an altitude rating chart (1,000-hour life, except for brushes) giving the safe allowable rating for the starter-generator. This chart shall be in the same range and general form or equivalent presentation as shown on Figure 5. The spacing and number of lines for pressure drop and temperature shall be so selected that the change in airflow and rating from one line to another shall be not more than 20 per cent.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16.1 Nameplate. The nameplate shall be irreversible and in conformance with Figure 6.

3.17 Workmanship.

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3.17.1 General. The starter-generator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.17.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.17.3 Cleaning. The starter-generator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs, sharp edges, and resin flash that may crumble shall be removed.

3.17.4 Safetying. All parts and connections shall be thoroughly secured to prevent accidental or vibratory loosening. Safety wire is not the preferred method.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of starter-generators shall be classified as follows:

a. Qualification testing (see 4.4)

b. Acceptance tests (see 4.5)

4.3 Test conditions.

4.3.1 Temperatures. Unless otherwise specified, all tests shall be conducted at a cooling air temperature within the range of 10°C to **40°C.** Ambient and cooling air temperatures shall not differ more than 5°C.

4.3.2 Brush life. All tests specified in 4.4.4 shall be performed on the starter-generator without replacement of any brushes.

4.4 Qualification testing.

4.4.1 Qualification test sample. The qualification test samples shall consist of three starter-generators. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

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4.4.2 Qualification test report. After the contractor completes the qualification tests, a qualification test report prepared in accordance with MIL-STD-831 shall be furnished to the qualifying activity.

4.4.3 Qualification test samples for the qualifying activity. Along with the qualification test report, the contractor shall submit a sample starter generator to the qualifying activity.

4.4.4 Qualification tests. Qualification tests shall consist of all tests described under 4.6 and as specified in table IV, conducted in any logical order to permit maximum utilization of test facilities:

TABLE IV. Qualification Tests

Test	No. 1	No. 2	No. 3
Examination of product	X	X	X
Maximum speed for regulation	X	X	X
Minimum speed for regulation	X	X	X
Heating, commutation, and minimum speed	X	X	
Overspeed	X	X	
No-load speed	X	X	X
Radial vibration	X	X	X
Paralleling voltage drop	X	X	X
Dielectric strength	X	X	X
Ripple voltage - generator	X	X	X
Cranking torque and speed	X	X	X
Radio interference		X	
Bonding	X	X	X
Radiated interference	X		
Endurance - starter	X		
Efficiency - generator	X	X	
Endurance - generator	X		
Consecutive cycle - starter	X	X	
Sea water immersion			X
Drive shear section	X		
Disassembly and inspection	X	X	
Bearing failure		X	
Altitude rating chart			X
Low temperature	X		
High temperature	X		
Humidity		X	
Vibration	X	X	
Fungus	X		
Explosion proof	X		
Low and high temperature storage		X	

4.5 Acceptance tests. Acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each starter-generator shall be subjected to the following tests, as described under 4.6:

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a. Examination of product.

b. Maximum speed for regulation.

c. Minimum speed for regulation.

d. No load speed.

e. Radial vibration.

f. Paralleling voltage drop.

g. Dielectric strength.

h. Ripple voltage.

i. Cranking torque and speed.

4.6 Test Methods.

4.6.1 Examination of products. The starter-generator shall be Inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Maximum speed for regulation. The starter-generator shall not be warmed up prior to this test. The generator shall be operated at 9,000 rpm and shall deliver rated voltage at no load with not more than 35 ohms in the external field circuit. (See 3.7.2)

4.6.3 Minimum speed for regulation. The generator shall be operated, and shall meet the requirements specified in 3.7.3.

4.6.4 Heating, commutation, and minimum speed. Provision shall be made for determining speed, terminal voltage, load current, field voltage, field current, and the resistance in series with the shunt field. Cooling air shall be supplied at the air inlet at the pressure specified in Table V. The temperature of the cooling air shall be determined by means of a temperature indicating device, the sensing element of which is located within the cooling air duct between the inlet of the generator and the pressure determining element. While the generator is cold, the resistance and temperature of the shunt field shall be determined for use in calculating the field temperature rise (average) during continuous operation at rated load. The generator shall reach its continuous operating condition when the rate of the shunt field temperature above the then existing ambient temperature does not increase more than l°C during a period of 5 minutes. The temperature attained will not cause rejection provided all other requirements of this specification are met. (See 3.7.4)

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TABLE V. Operating Conditions

Operating Time	Cooling Air Pressure Drop Across Generator (inces of water)	Drive Speed (rpm)	Terminal Voltage	Load Current (amp)	Altitude (feet)
Continuous	12	3,800	26	150	Sea level
Continuous	12	4,400	30	200	Sea level
Continuous	12	8,000	30	200	Sea level
Continuous	12	8,100	30	200	Sea level
5 minutes	-	10,100	-	-	Sea level
Continuous	* 6	8,000	30	165	60,000

* Cooling air temperature -50°C to -60°C for high altitude tests.

4.6.4.1 Heating. The ability of the starter-generator to deliver 200 amp at 26V and 4400 rpm shall-be determined.

4.6.4.2 Commutation. Immediately following the heat run with the starter-generator hot, the commutation of the starter-generator shall be observed over the speed range for no-load, half-load, and rated-load. There shall be no more than fine, pin-point sparking of the brushes during this test.

4.6.4.3 Minimum speed. At no time during the heat run at minimum speed of 3,800 rpm shall the required resistance external to the shunt field be less than 1.25 ohms.

4.6.5 Overspeed. This test shall be made while the starter-generator is hot as a result of testing, and shall be made at no-load with the shunt field circuit open and at an overspeed of 10,000 rpm. The starter-generator shall demonstrate ability to operate under overspeed conditions for 5 minutes without mechanical failure, throwing of solder or varnish, or impairing of electrical performance. (See 3.7.5)

4.6.6 No-lead speed. At zero torque, the starter no-load speed shall be not less than 2,500 rpm with 23V dc applied at the terminals. (See 3.7.6)

4.6.7 Radial Vibration. After completion of the preceding tests, the radial vibration shall not exceed 0.001 inch total deflection at maximum rated speed when the starter-generator is operated as a starter. It shall be permissible to employ a drive shaft support during the test. (See 3.7.7)

4.6.8 Paralleling voltage drop. The generator shall be tested for and shall meet the paralleling voltage drop requirements of 3.7.8.

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4.6.9 Dielectric strength. While the starter-generator is hot as a result of testing, it shall withstand the following test voltage at commercial frequency, applied between windings and between winding and frame, for the specified time: (See 3.7.9)

500V root mean square (ins) for 1 minute or
600V rms for 1 second

Radio noise filters may be removed or disconnected for this test.

4.6.10 Ripple voltage-generator. Ripple voltage shall be determined by means of a peak-reading vacuum tube voltmeter in series with a 4.0 microfarad capacitor as shown on Figure 4. The generator shall be operated at speeds of 3,800, 6,000, and 8,100 rpm under conditions of no-load, half-load and rated-load. A manually operated field rheostat shall be used in lieu of a voltage regulator. Peak-voltage readings shall be taken with the voltmeter successively connected for each of the two polarities and the higher of the two readings shall be not more than 1.5V. The voltmeter shall be an approved type. At no time during this test shall a battery be used. (See 3.7.10)

4.6.11 Cranking torque and speed. The starter shall be tested to determine compliance with the minimum cranking torque and speed requirements specified on Figures 1 and 2. An external resistance of 1.25 ohms shall be connected in the shunt field circuit while conducting starter tests. (See 3.7.11)

4.6.12 Radio interference.

4.6.12.1 Conducted radio interference measurements. Unless otherwise specified, conducted radio interference measurements shall be made between each load of the starter-generator and negative grounded. The measurement shall be made at the point of entry of leads to each unit, With a filter mounted and properly connected to terminals A, B, and C, the conducted radio noise voltage shall not exceed 2,000 microvolt (UV) over the 0.15 to 20.0 megacycles (me) frequency range when operated at speeds ranging from 3,800 to 8,100 rpm under conditions of no-load and rated-load. (See 3.9)

4.6.12.2 Noise meters. The standard instruments for making radio interference measurements (0.15 to 20.0 mc) shall be in accordance with MIL-STD-826.

4.6.12.3 Bonding. The direct-current resistance between the commutator end bell and face of mounting flange shall be not greater than 0.001 ohm.

4.6.12.4 Radiated interference. Radiated interference measurements in accordance with MIL-STD-826 shall be conducted over the frequency range 0.15 to 1,000 mc.

4.6.13 Endurance-starter. The starter shall complete a 600-cycle endurance test, as specified herein without damage to, or replacement of, component parts. Each cycle shall consist of three consecutive phases, as shown in Table III.

4.6.13.1 After completion of this endurance test, the peak efficiency of the starter shall not have decreased more than 5 per cent from the original value. Total brush wear shall not have exceeded 20 per cent of the original effective brush length on any one brush.

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4.6.14 Efficiency-generator. The generator shall be subjected to the operating conditions specified in Table V. The efficiency of the generator shall be determined at rated terminal voltage, and the speed and load conditions tabulated in Table V. The efficiency shall be not less than 60 per cent under any of the conditions specified. (See 3.7.13)

4.6.15 Endurance-generator. The endurance test of the generator shall consist of eight cycles of runs No. 1 through 3, followed by a single run, No. 4, as specified in Table II without damage to, or replacement of, component parts. Upon completion of the test, projected brush life shall be not less than 500 hours.

4.6.16 Consecutive cycle-starter. The starter-generator shall be operated for three consecutive endurance cycles, as shown in Table III, at 30-second intervals without damage. Cooling air shall not be applied during this test.

4.6.17 Sea water immersion. The starter-generator shall be subjected to sea water immersion in accordance with 3.7.16.

4.6.18 Drive shear section. Sufficient torsional force shall be applied to the drive shear section to result in Its failure. Failure shall occur within the limits specified in 3.7.17.

4.6.19 Disassembly and inspection. The starter-generator shall be disassembled and inspected. There shall be no evidence of major deterioration, corrosion, or undue wear.

4.6.20 Bearing failure. After disassembly and inspection, the starter-generator shall be reassembled minus the lubricant, cage, and seals in the bearing located in the flange end of-the unit. The starter-generator shall be operated as a generator delivering 200 amp 30V dc, 8,000 rpm with a cooling air-pressure drop across the generator equivalent to 12 inches of water. The starter-generator shall be operated for a period of not less than 4 hours. During this test, only intermittent sparks that emanate from within shall be permitted. Any tendency to flame shall be cause for rejection.

4.6.21 Altitude rating chart. The altitude rating chart specified in 3.15 shall be justified by tests. Weight of airflow shall be shown at 4,400, 6,000, and 8,000 rpm at varying air density (altitude) and pressure drop across the generator up to and including 40 inches of water. The output current shall be shown at 4,400, 6,000, and 8,000 rpm.

* 4.6.22 Environmental. The starter-generator shall be subjected to the following environmental tests in accordance with the specified procedure of MIL-STD-810.

4.6.22.1 Low temperature. Low temperature test shall be in accordance with procedure I for a minimum of 72 hours. At the end of this test, the starter-generator shall complete without damage, three consecutive cycles in accordance with Table III.

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4.6.22.2 High temperature. The high-temperature test shall be in accordance with procedure I for a minimum of 8 hours. At the end of this test the starter-generator shall complete, without damage, three consecutive cycles in accordance with Table III. External cooling shall not be permitted during this test.

4.6.22.3 Humidity. The humidity test shall be in accordance with procedure I. At the end of this test the starter-generator shall operate satisfactorily as a generator at full-rated load for a period of 2 hours.

4.6.22.4 Vibration. For resonance, the vibration test shall be in accordance with procedure VII. For endurance, the test shall be in accordance with procedure VIII or XII, except the applied vibration as monitored on the test fixture shall be reduced wherever necessary to prevent the vibratory acceleration from exceeding -40 times standard gravity constant (where standard gravity constant = g) at any point on the basic generator frame. The resonant frequency shall be not less than 200 cycles per second (cps). Any mechanical damage due to vibration tests shall be cause for rejection.

4.6.22.5 Fungus. The fungus test shall be in accordance with procedure 1. Immediately following this test, the starter-generator shall operate satisfactorily as a generator at full-rated load for a period of 2 hours.

4.6.22.6 Explosion proof. The explosion proof test shall be in accordance with procedure I. Any ignition of ambient explosive gaseous mixture shall be cause for rejection.

4.6.22.7 Low-and high-temperature storage. The starter-generator shall be soaked for a period of 8 hours at -62°C and 8 hours at +93°C. In each case, the starter-generator shall be returned to room temperature and satisfactorily operated as a generator at full load for 2 hours.

4.7 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

* 5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C or Commercial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, each starter-generator shall be individually preserved and packaged in accordance with MIL-P-116, Method II.

* 5.1.2 Level C. The Level C preservation for the starter-generator shall conform to the MIL-STD-794 requirements for this level.

* 5.1.3 Commercial. The commercial preservation of starter generator shall be in accordance with the requirements of ASTM D 3951-82.

* 5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.2).

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5.2.1 Level A. Starter-generators preserved and packaged as specified in 5.1.1 shall be palletized, 20 each per pallet in accordance with MIL-STD-147. Quantities of 19 each or less shall be packed in export type shipping containers conforming to PPP-B-601. As far as practicable, exterior shipping containers shall be of uniform shape and size, and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the standard or appendix of the specification as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Starter generators preserved and packaged, as specified in 5.1.1, not to be overboxed for domestic shipment. Quantities of 20 each or more shall be palletized in accordance with MIL-STD-147. For quantities of 19 each or less shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional. Closure and strapping shall be in accordance with the standard or appendix of the specification as applicable.

5.2.3 Level C. The Level C packing for starter-generator shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Commercial. The preserved starter-generator shall be packed in accordance with the requirements of ASTM D 39s1-82.

5.2.5 Palletization. When specified (see 6.2) palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3 Physical protection. Cushioning, blocking, bracing and bolting shall be in accordance with MIL-STD-1186, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of MIL-STD-1186 shall be waived when preservation, packaging and packing of the item is for immediate use or when drop tests of MIL-P-116 are applicable.

* 5.4 Marking.

* 5.4.1 Levels A, B, and C. In addition to any special or other identification marking, required by the contract (see 6.2), each unit pack, intermediate and exterior container and palletized load shall be marked in accordance with MIL-STD-129.

* 5.4.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM-D-3951-82.

6. NOTES

6.1 Intended use. The starter-generator covered by this specification is intended for use in starting gas turbine type engines, and to provide electric power to the aircraft.

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* 6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title. number, and date of this specification.

b. When the torsional vibration damper is required to be furnished (see 3.4.5).

c. Invitation for bids, contracts, and purchase orders shall specify the applicable levels of preservation, packaging, and packing required.

6.3 Qualification. With respect to products requiring qualification, awards will-be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is San Antonio ALC/MMEDO, Kelly APB, Texas 78241 and information pertaining to qualification of products may be obtained from that activity.

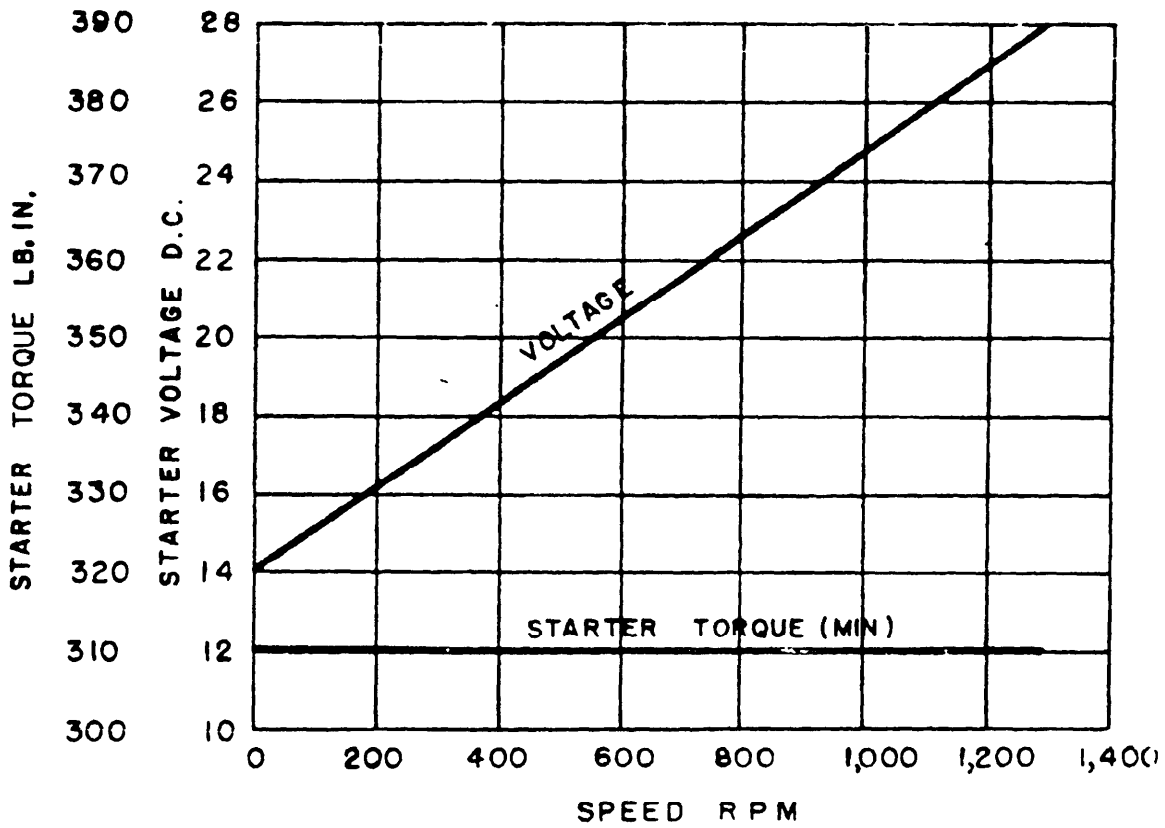
6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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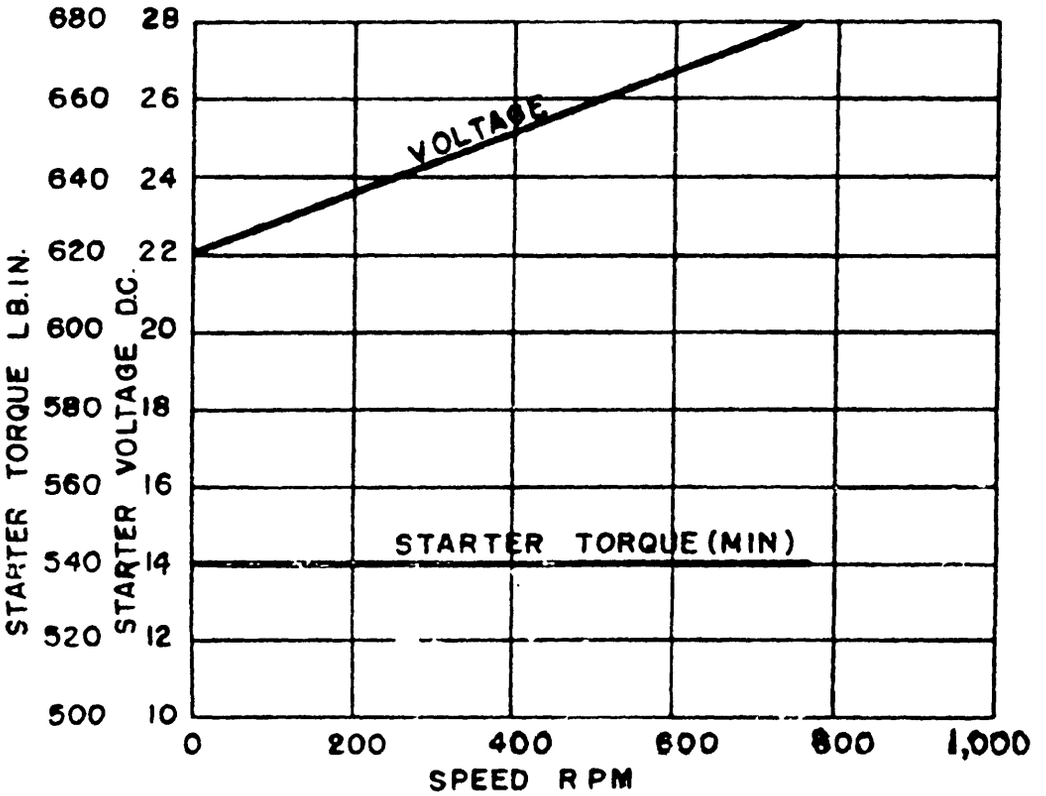
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SPEED-TORQUE CHARACTERISTICS
STARTER-CURRENT-500 AMPERES
FULL FIELD WITH 1.25 OHMS EXTERNAL
RESISTANCE IN SHUNT FIELD CIRCUIT
TEMPERATURE OF WINDINGS STABILIZED @ 25°C

FIGURE 1. STARTER GENERATOR CHARACTERISTICS, 500 AMPERES

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SPEED-TORQUE CHARACTERISTICS
STARTER CURRENT-800 AMPERES
FULL FIELD WITH 1.25 OHMS
EXTERNAL
RESISTANCE IN SHUNT FIELD
CIRCUIT
TEMPERATURE OF WINDINGS
STABILIZED @ 25°C

FIGURE 2. STARTER GENERATOR CHARACTERISTICS, 800 AMPERES

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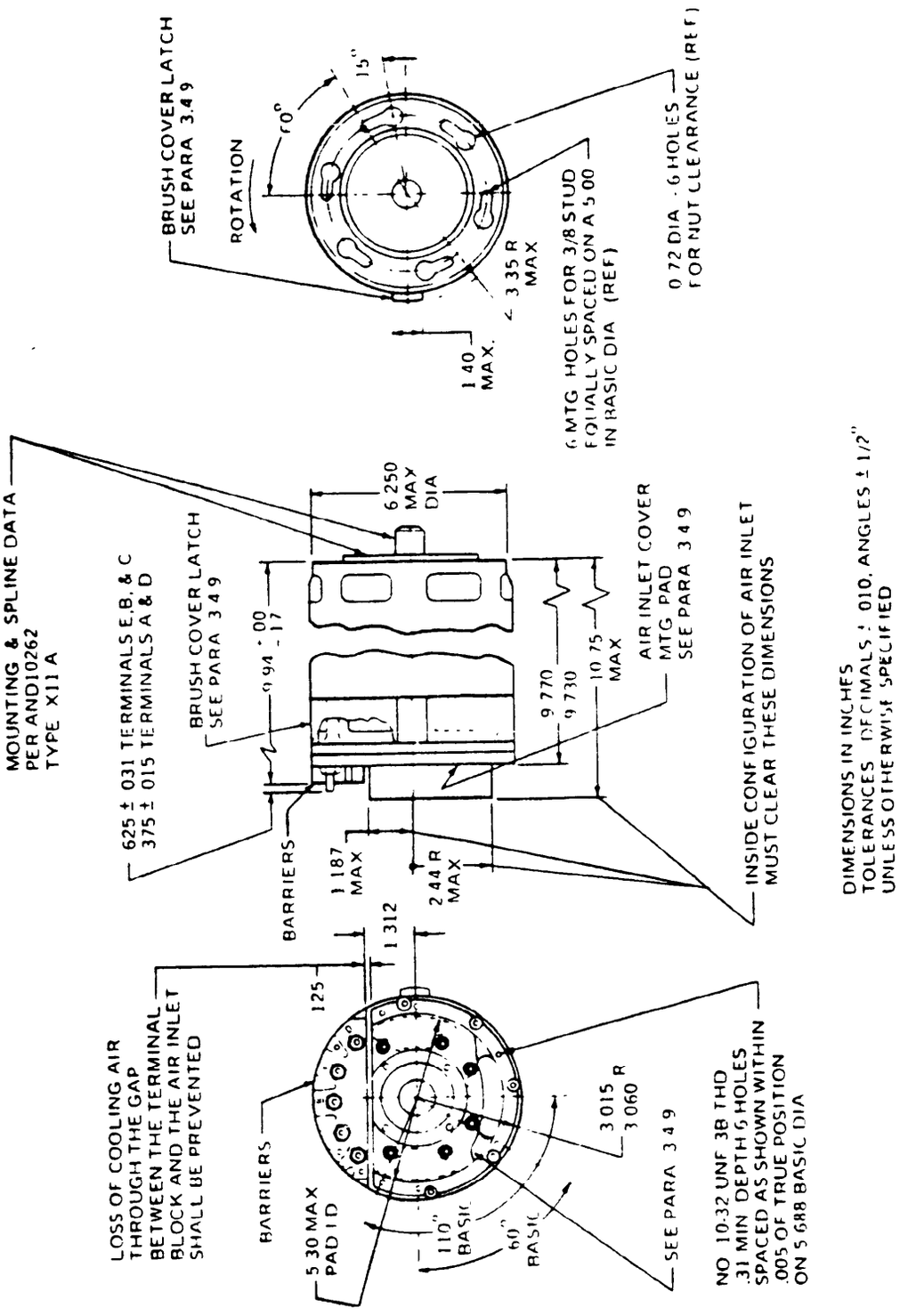


FIGURE 3. STARTER-GENERATOR

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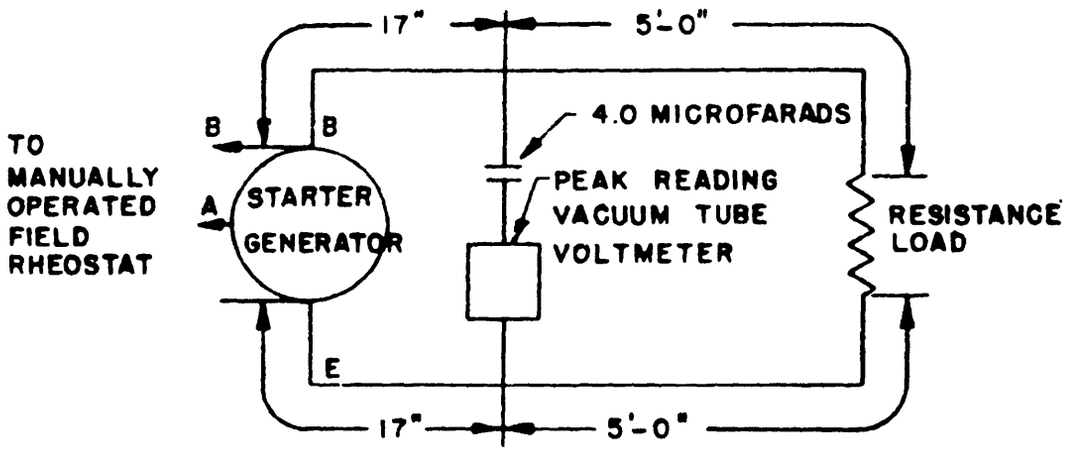
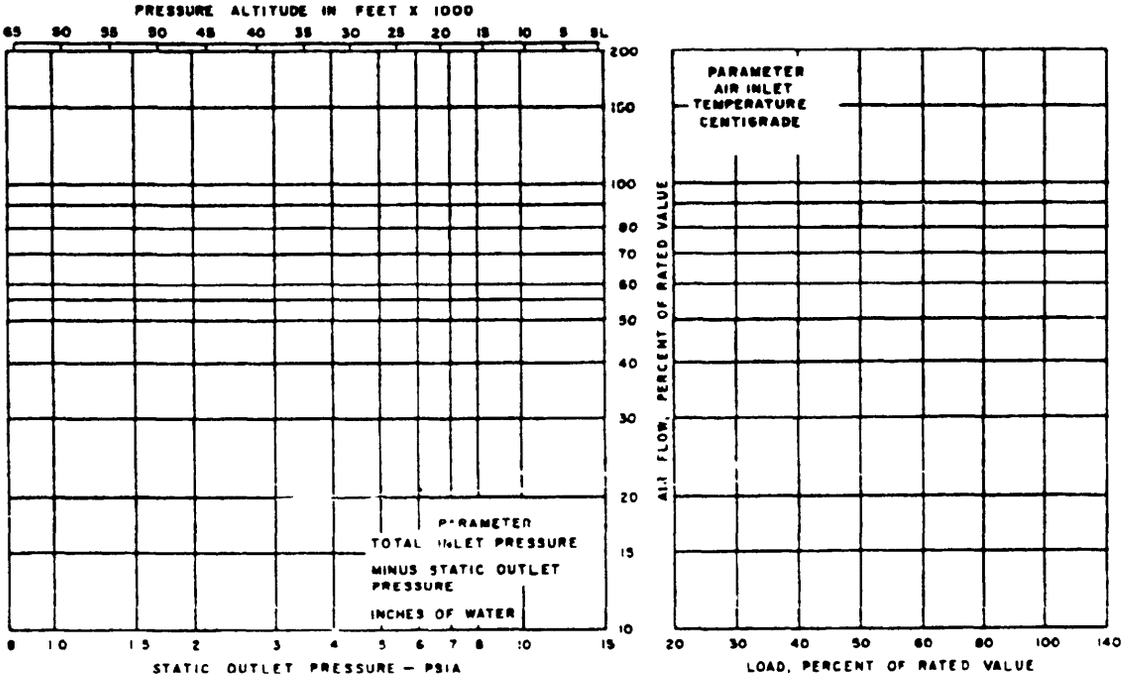


FIGURE 6. RIPPLE VOLTAGE TEST SET-UP

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GENERATOR RATING CHART

MANUFACTURER

MANUFACTURER P/N — ASSY DWG NO

SPECIFICATION NO — DWG NO

AIR INLET CONFIGURATION

RATED LOAD AT VOLTS

SPEED

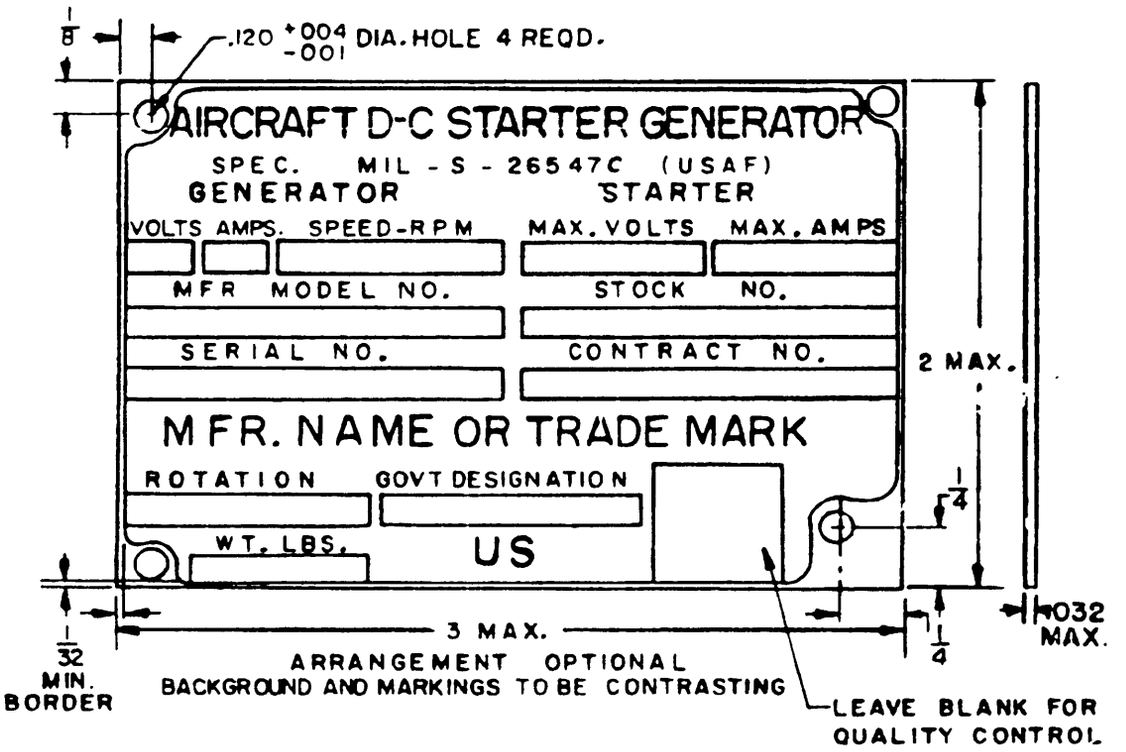
RATED AIRFLOW LB / MIN

OTHER DATA AND REPORTS

NOTE THIS FORM NOT AVAILABLE THROUGH SUPPLY CHANNELS, MUST BE REPRODUCED LOCALLY

FIGURE 5 STANDARD FORM OF GENERATOR RATING CHART

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DIMENSION IN INCHES
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FIGURE 6. NAMEPLATE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-26547C	STARTER-GENERATOR ENGINE STU-6A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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